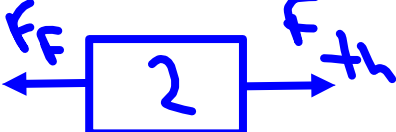
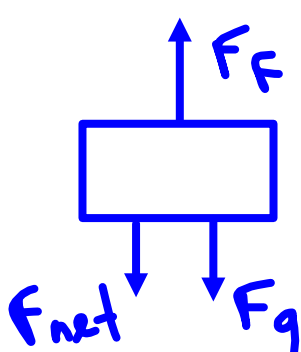


# Forces Unit Review

1 a)   $F_{net} = ma = F_{th} - F_F$   
 $2(4) = F_{th} - 10$   
 $F_{th} = 18 \text{ N}$

b)   $F_{net} = ma = F_g - F_F$   
 $(0.5)(8.5) = 0.5(9.8) - F_F$   
 $F_F = 0.65 \text{ N}$

c)  $F_{net} = ma \Rightarrow v_f = v_o + at$   
 $14.7 = 0 + a(6)$   
 $a = 2.5 \text{ m/s}^2$   
 $0.049 = m(2.5)$   
 $m = 0.02 \text{ kg} = 20 \text{ g}$

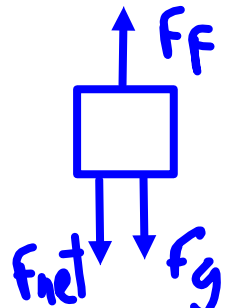
d)  $F_{net} = ma$   
 $450 = 120a$   
 $a = 3.8 \text{ m/s}^2$

$$d = v_o t + \frac{1}{2}at^2 = \frac{1}{2}(3.8)(5)^2 = 47 \text{ m}$$

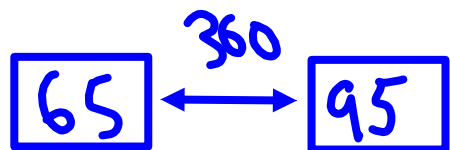
e)  $v_f = v_o + at$   
 $4 = a(6)$   
 $a = 0.67 \text{ m/s}^2$   
 $F_{net} = ma = 24(0.67) = 16 \text{ N}$

f)  $v_f = v_o + at$   
 $33 = 15 + a(25)$   
 $a = 7.2 \text{ m/s}^2 \Rightarrow F_{net} = ma$   
 $4500 = m(7.2) \Rightarrow m = 625 \text{ kg}$

g)  $v_f = v_o + at$   
 $46 = 21 + a(0.7)$   
 $a = 36 \text{ m/s}^2 \Rightarrow F_{net} = ma = 0.5 \text{ kg}(36) = 18 \text{ N}$

h)   $F_{net} = ma = F_g - F_F = mg - F_F$   
 $2a = 2(9.8) - 5$   
 $a = 7.3 \text{ m/s}^2$   
 $v_f = v_o + at = 0 + 7.3(6) = 44 \text{ m/s}$

2

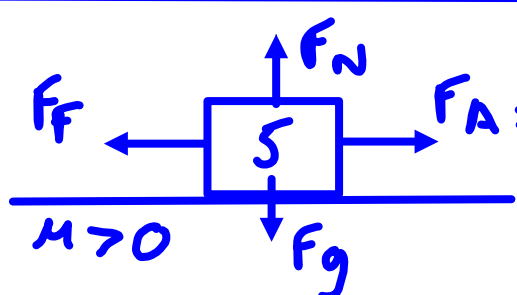


$$F_{\text{net}} = ma$$

$$360 = 65a \quad \text{vs.} \quad 360 = 95a$$

$$a_{65} = 5.5 \text{ m/s}^2 \leftarrow \quad \text{vs.} \quad a_{95} = 3.8 \text{ m/s}^2 \rightarrow$$

3



$$\text{a) } F_{\text{net}} = ma = 5(4) = \boxed{20 \text{ N}}$$

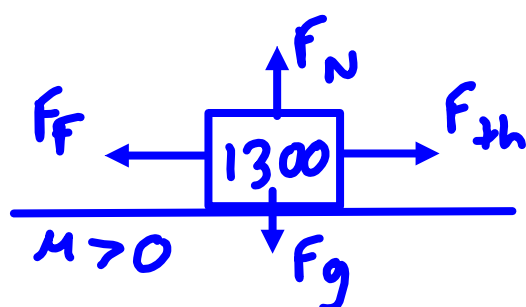
$$\text{b) } F_{\text{net}} = F_A - F_f \Rightarrow 20 = 30 - F_f$$

$$\boxed{F_f = 10 \text{ N}}$$

$$\text{c) } F_f = \mu F_N \rightarrow mg$$

$$10 = \mu(5)(9.8) \Rightarrow \boxed{\mu = 0.2}$$

4



$$\text{a) motion: } v_f^2 = v_i^2 + 2ad$$

$$0^2 = 60^2 + 2a(325)$$

$$\boxed{a = 55 \text{ m/s}^2 \text{ [backward]}}$$

$$\text{b) forces. } ma = F_f - F_{th}$$

$$1300(55) = 0.65(1300)(9.8)$$

$$\boxed{F_{th} = 1200 \text{ N}}$$

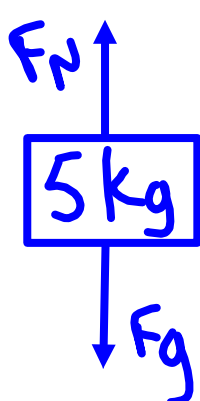
5

$$\text{a) } -23 = 45(x_f - 0.75) \Rightarrow \boxed{x_f = 24 \text{ cm}}$$

$$\text{b) } 14 = 63(x_f - 0.25) \Rightarrow \boxed{x_f = 47 \text{ cm}}$$

6

$$\text{*at rest } F_{\text{net}} = 0$$



$$F_N + F_{g3.2} = F_{g5}$$

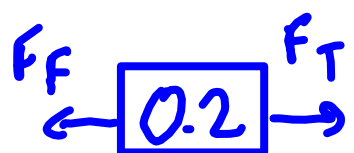
$$F_N + 3.2(9.8) = 5(9.8)$$

$$\boxed{F_N = 18 \text{ N}}$$

$$7 \quad (m_1 + m_2)a = F_A - F_f \rightarrow \mu(m_1 + m_2)g$$

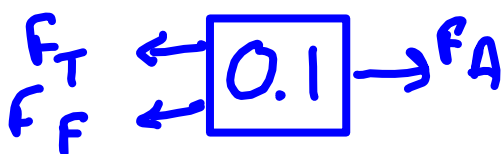
$$(0.2 + 0.1)4.4 = F_A - 0.35(0.2 + 0.1)(9.8)$$

$$\boxed{F_A = 2.3 \text{ N}}$$



$$ma = F_T - F_f$$

$$0.2(4.4) = F_T - 0.35(0.2)(9.8)$$



$$ma = F_A - F_T - F_f$$

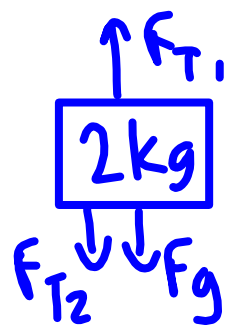
$$(0.1)(4.4) = 2.3 - F_T - 0.35(0.1)(9.8)$$

$$\boxed{F_T = 1.6 \text{ N}}$$

$$8. \quad ma = F_{g2/3} - F_{g1}$$

$$(1 + 2 + 3)a = (2 + 3)9.8 - 1(9.8)$$

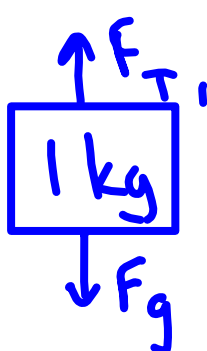
$$\boxed{a = 6.5 \text{ m/s}^2 \downarrow}$$



$$ma = F_{T2} + F_g - F_{T1}$$

$$2(6.5) = F_{T2} + 2(9.8) - 16$$

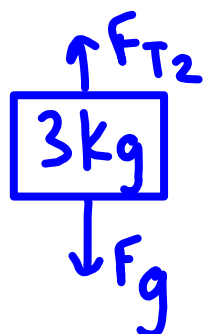
$$\boxed{F_{T2} = 10 \text{ N}}$$



$$ma = F_{T1} - F_g$$

$$1(6.5) = F_{T1} - 1(9.8)$$

$$\boxed{F_{T1} = 16 \text{ N}}$$



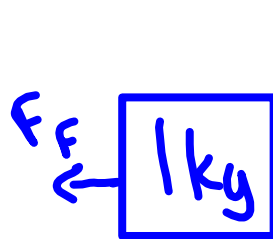
$$ma = F_g - F_{T2}$$

$$3(6.5) = 3(9.8) - F_{T2}$$

$$9 \quad ma = F_g - F_f \rightarrow \mu F_N \rightarrow \mu mg$$

$$(1 + 3 + 2)a = 2(9.8) - 0.15(1 + 3)(9.8)$$

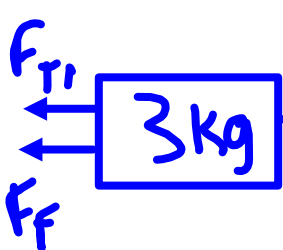
$$\boxed{a = 2.3 \text{ m/s}^2 \downarrow}$$



$$ma = F_{T1} - F_f \rightarrow \mu F_N \rightarrow \mu mg$$

$$1(2.3) = F_{T1} - 0.15(1)(9.8)$$

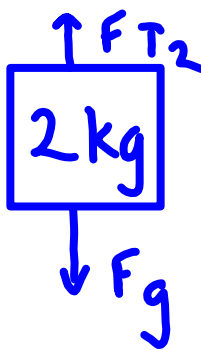
$$\boxed{F_{T1} = 3.8 \text{ N}}$$



$$ma = F_{T2} - F_{T1} - F_f \rightarrow \mu mg$$

$$3(2.3) = F_{T2} - 3.8 - 0.15(3)(9.8)$$

$$\boxed{F_{T2} = 15 \text{ N}}$$

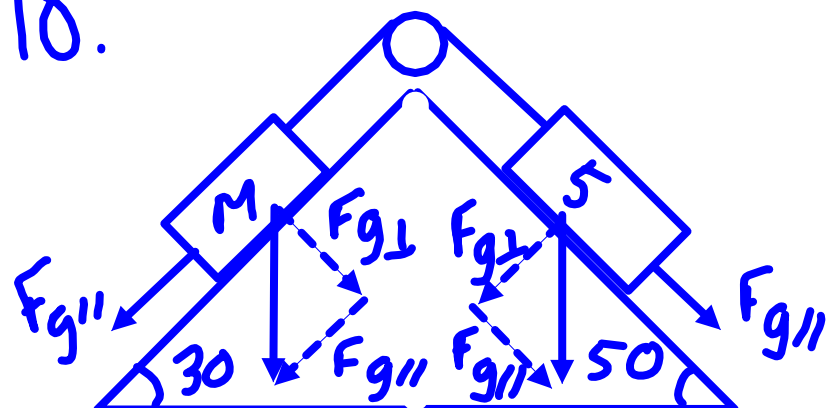


$$ma = F_g - F_{T2}$$

$$2(2.3) = 2(9.8) - 15$$

$$4.57 = 4.57 \checkmark$$

10.

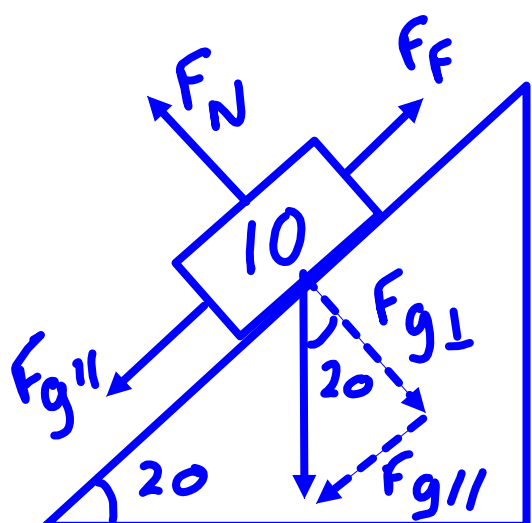


$$F_{g||5} = F_{g||M}$$

$$5(9.8)\sin 50 = M(9.8)\sin 30$$

$$M = 766 \text{ kg}$$

11.



$$a) F_N = F_{g\perp} = mg \cos \theta$$

$$F_N = 10(9.8) \cos 20 = 92 \text{ N}$$

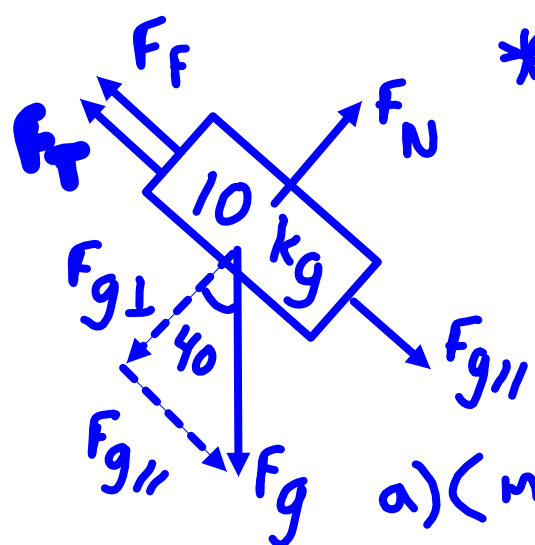
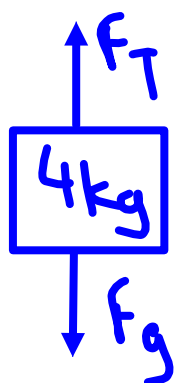
$$b) F_F = F_{g||} = mg \sin \theta = 10(9.8) \sin 20 = 34 \text{ N}$$

(not moving)

$$c) F_F = \mu F_N \Rightarrow 34 \text{ N} = \mu(92 \text{ N})$$

$$\mu = 0.36$$

12

\*  $F_{g||}$  vs  $F_g$ 

$$10(9.8) \sin 40 \text{ vs. } 4(9.8)$$

$$63 \text{ N} > 39 \text{ N} \quad (F_F \uparrow)$$

$$a) (m_1 + m_2)a = m_2 g \sin \theta - \mu m_2 g \cos \theta - m_1 g$$

$$(4 + 10)a = 10(9.8) \sin 40 - 7.5 - 4(9.8)$$

$$a = 1.16 \text{ m/s}^2 \searrow$$

$$m_1 a = F_T - F_g$$

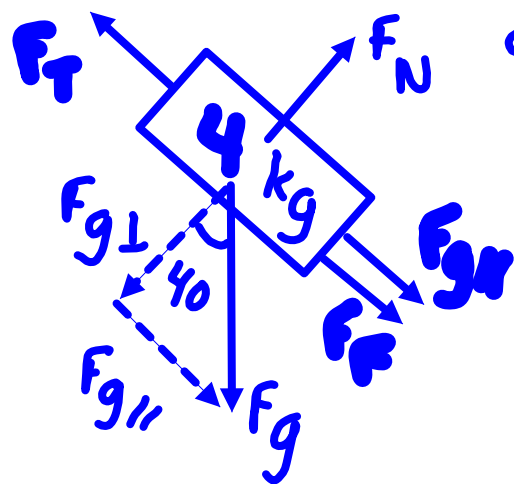
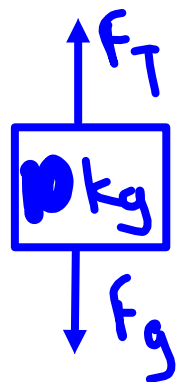
$$4(1.2) = F_T - 4(9.8)$$

$$F_T = 43.9 \text{ N}$$

$$m_2 a = F_{g||} - F_F - F_T$$

$$10(1.2) = 10(9.8) \sin 40 - 0.1(10)(9.8) \cos 40$$

13



$$a) (m_1 + m_2)a = F_{g||} - F_{g||2} - F_F$$

$$(m_1 + m_2)a = m_1 g - m_2 g \sin \theta - \mu m_2 g \cos \theta$$

$$(10 + 4)4.53 = 10(9.8) - 4(9.8) \sin 40 - \mu(4)(9.8) \cos 40$$

$$\mu = 0.312$$

$$b) m_1 a = F_g - F_T$$

$$10(4.53) = 10(9.8) - F_T$$

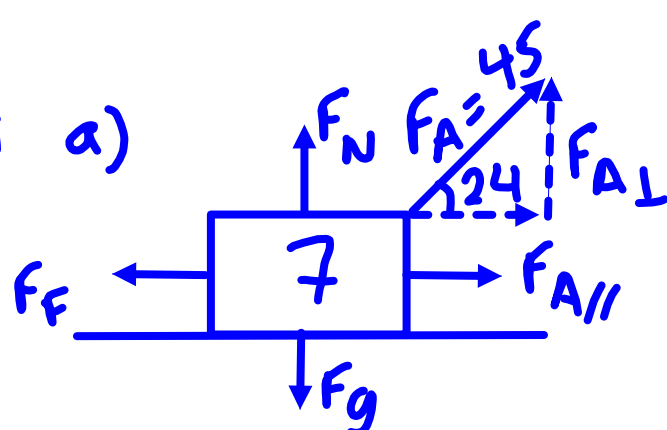
$$m_2 a = F_T - F_{g||} - F_F$$

$$4(4.53) = F_T - 4(9.8) \sin 40$$

$$- 0.312(4)(9.8) \cos 40$$

$$F_T = 52.7 \text{ N}$$

14 a)



$$F_{\text{net}} = ma = F_{A||} - F_F \rightarrow \mu F_N$$

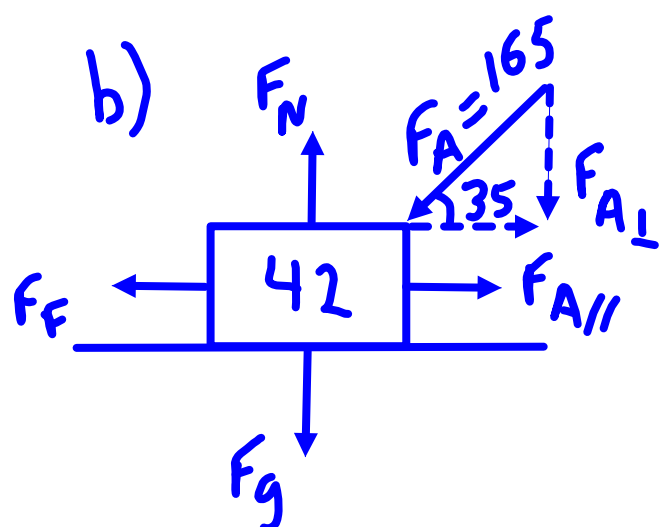
$$7a = 45 \cos 24 - 0.52 [7(9.8) - 45 \sin 24]$$

$$a = 2.1 \text{ m/s}^2 \rightarrow$$

$$F_N + F_{A\perp} = F_g \Rightarrow F_N = F_g - F_{A\perp}$$


---

b)



$$F_N = F_{A\perp} + F_g$$

$$F_{\text{net}} = ma = F_{A||} - F_F \rightarrow \mu F_N$$

$$42(0.95) = 165 \cos 35 - \mu [165 \sin 35 + 42(9.8)]$$

$$\mu = 0.19$$